

Mathieu Néron

SOFTWARE ENGINEER

San Francisco, CA

■ (628) 358-8813 | ✉ mathieu.neron.ing@gmail.com | 📱 mathieu-neron | 🌐 mathieu-néron-b4419522

Summary

Software engineer with 15+ years shipping at scale. Currently building fraud-risk infrastructure at Highnote, processing **3M+ monthly card transactions** (~\$350M+ in monthly settlement volume); previously Senior → LMTS at Tableau, building authoring and server features used by **87K+ customer companies** and **3M+ Tableau Public creators**. Background spans fintech, data visualization, and e-signature; actively exploring applied AI/ML through side projects, with recent focus on distributed-systems architecture (DDIA, Kafka, batch / stream processing).

Experience

Highnote

San Francisco, CA

SENIOR SOFTWARE ENGINEER

Aug. 2023 – Present

- *Java, TypeScript, GraphQL, GCP, Spanner, BigQuery, Temporal, Elasticsearch, Python, ANTLR*
- Owned **OFAC sanctions screening** on Highnote's synchronous money-movement critical path (Visa Direct / RTP / ACH); landed a three-layer optimization — in-memory early-exit, False/True positive lookups deferred to candidate matches, Spanner bounded-staleness reads — to hit a **P99 ≤ 100 ms** SLA for a major customer migration; **99.996%** of ~244K monthly screenings now bypass the database entirely.
- Designed and shipped a customizable **geofencing rule** now deployed across **7 subscribers** (19 rule instances) on Highnote's rule engine; mid-2025 tuning lifted true-positive rate **3–4×** on the Fleet-program enforcement path.
- Led migration of the core SMS stack from AWS Pinpoint to **Twilio** and built the platform's first **bi-directional SMS fraud-detection workflow** on **Temporal** — a real-time-risk differentiator anchoring a flagship enterprise rollout and central to the platform's pitch to new fraud-sensitive subscribers.
- Drove adoption of **Temporal** (durable workflows) and **Elasticsearch** as first production use cases at Highnote; both now standard across multiple teams.
- Extended the **ANTLR grammar** powering Highnote's natural-language query layer with fuzzy / typo-tolerant search.

Tableau (Salesforce)

Palo Alto, CA

SENIOR SOFTWARE ENGINEER → LMTS

Jun. 2019 – Jul. 2023

- *React, Redux, TypeScript, Java, C++, Qt* — Promoted to Lead Member of Technical Staff in 2020.
- Onboarded a new AI-predictive-model content type in Tableau Server for **Salesforce Einstein Discovery** integration — contributing to the server stack delivering **80B+ AI-powered predictions per day**.
- Drove the **Object Graph** view in Tableau's Data pane from beta to GA — the default relationships editor in Tableau's multi-table data model, part of the authoring stack behind Tableau Public's **7M+ vizzes** from **3M+ creators**; shipped transition animations, tooltips, sorting, and re-rooting.
- Contributed to the **Data Preview Area** hybridization — metadata grid plus embedded relationship editor in a single surface.

Paysafe

Montréal, Canada

JAVA DEVELOPER

2016 – 2019

- *Angular 2+, TypeScript, Java, Spring, Oracle*
- Shipped the new Angular 2+ **Merchant Portal** frontend from greenfield.
- **Cut frontend test runtime 6×** by migrating from PhantomJS to Nightmare.

eSignLive

Montréal, Canada

JAVA DEVELOPER

2014 – 2016

- *Java, Spring, Guava, AWS, MySQL, Jenkins*
- **Cut Windows document converter downtime by 50%**.
- Integrated Equifax, Twilio, PDFBox, and Google Noto Fonts.

Avanade

Montréal, Canada

APPLICATION DEVELOPER

2012 – 2014

- Enterprise web engineering for Canadian National Railway (*www.cn.ca* redesign on Sitecore / ASP.NET) and Hydro-Québec (internal time-tracking app; led frontend-stack selection — SAPUI5 + Infragistics).

Projects

github.com/mathieu-neron/RealTube

Personal project

REALTUBE

2026

- Browser extension (Chrome/Firefox, Manifest V3) for crowdsourced detection and hiding of AI-generated YouTube content across feed, search, recommendations, and Shorts.
- Architected a dual-backend split — **Go** (Fiber v3) on the hot path, **Python + FastAPI** for ML-ready services — to isolate low-latency endpoints from future inference workloads; **TypeScript + React 19** extension, PostgreSQL + Redis + Docker + Prometheus.
- Privacy-engineered hash-prefix lookup with **5000-round SHA256 stretching** to harden against brute-force reverse lookups; contract-tested API, e2e tests, 10 architecture design docs.

Skills

Languages Java, TypeScript/JavaScript, Python, Go, C/C++, C#

Frameworks Spring, Temporal, Node.js, React/Redux, Angular, GraphQL, REST, Qt, .NET, ANTLR

Platforms GCP, AWS, Spanner, BigQuery, PostgreSQL, Elasticsearch, Redis, Oracle, MySQL, Docker

Spoken French (native), English (fluent)

Education

Université de Sherbrooke

Sherbrooke, Canada

B.ENG., COMPUTER ENGINEERING

2007 – 2011

- Specialization in Software Engineering and Artificial Intelligence.

Internships

Electronic Arts Canada

Burnaby, Canada

SOFTWARE ENGINEER INTERN

Summer 2011

- Built a distributed C++ / C# load-test harness running thousands of concurrent bots against **SSX**'s multiplayer server; used internally and demoed at **Gamescom 2011**.

Autodesk | Ubisoft | Lockheed Martin

Canada

EARLIER INTERNSHIPS

2008 – 2010

- **Autodesk** (2010) — animation plug-in inside Maya (C++ / Qt). **Ubisoft** (2009) — QA automation (C# / .NET). **Lockheed Martin** (2008) — software evaluation.